

Appendix Documents

5. Comprehensive Pavement Management Plan

TOWN OF LUNENBURG
DEPARTMENT OF PUBLIC WORKS



PAVEMENT MANAGEMENT PLAN

PAVEMENT MANAGEMENT PLAN

At the request of the Board of Selectmen the Lunenburg Department of Public Works is pleased to present this Pavement Management Plan (PMP). This report details the reasons for undertaking a PMP, the process of data collection, how street repairs are prioritized, and other factors associated with maintenance of paved roads in Lunenburg.

Background on Roadway Infrastructure

Over the years, the Town has made a significant financial investment in its infrastructure. This infrastructure includes school and municipal buildings and the multi-year sewer project. Often overlooked but no less important, the roadway network is part of the Town's overall infrastructure. Like any piece of the infrastructure, the roadway network must be maintained in order to protect the public's investment. Deferring maintenance on the road system will only increase costs over time as the condition of the asset further deteriorates.

The Town maintains approximately 100 miles of roads. Current road construction estimates put the cost of reconstruction at approximately \$700,000 per road mile. With 100 miles of road, this asset has a value of about \$70 million. Unfortunately, the Town's financial resources for maintenance are limited to Chapter 90 State assistance and locally budgeted funds. Chapter 90 provides about \$320,000 per year and the FY10 budget commits \$70,000 for road improvements tools and materials. It should be noted that the General Highway Maintenance line within the departmental budget has been drastically reduced, often times mid-year due to reductions in local aid further postponing necessary maintenance.

Like many communities, the Town has not kept pace with the maintenance of its roads. Competing interests like public education, public safety, and general government operations often take priority over roadway improvements. Roadway improvements prioritized according to the squeaky wheel principle have been the norm, but is not a prudent, cost effective, or objective approach to maintaining such a valuable asset.

Furthermore, continued deferred maintenance will cost the Town dearly in the future. The cost of bituminous concrete (asphalt) has increased approximately 2 ½ times over the past 10 years. Although some costs may be slightly lower due to the current economic slump, there is no reason to believe paving costs will decline in the future.

The Board of Selectmen clearly understands the need to protect the roadway infrastructure and invest in its maintenance. For this reason the Board instructed staff to prepare a pavement management plan. As the Board rightly acknowledges, this investment needs to be done as efficiently and cost effectively as possible and it must be done in a systematic fashion. A Pavement Management System is the best way to facilitate this.

Pavement Management System

A Pavement Management System provides an objective way of programming roadway improvements with the goal of maximizing available resources and extending the life of the pavement system. The system establishes a baseline condition of roadways which is used to

establish a list of prioritized improvement recommendations. Roadway segments are prioritized according to a formula that includes the following:

- Functional Classification
- Pavement Defect Value

How the plan was prepared

The first step in preparing the PMP was to evaluate the current condition of each road or road segment. Longer roads are broken into segments reflecting past paving history and to facilitate manageable evaluation. A survey form was created to collect data on each roadway segment. The data on the survey form was then used to establish a priority ranking.

The public works industry uses two primary factors when evaluating pavement conditions: functional classification and pavement defects. Each is discussed below:

Functional classification: Roads are classified in a hierarchy according to how they are used. The functional classification is primarily a reflection of the volume of traffic a road accommodates. For the purposes of this PMP the following classes of roads are identified:

- **Main Road**: Typically contains high traffic volume and either permits traffic flow from one main road to another or acts as an inter-town connector. Example: Lancaster Avenue.
- **Secondary Main Road**: Typically contains higher traffic volume than either a collector road or a local road with less volume than a main road, also acting as a connector between main roads. Example: Sunny Hill Road.
- **Collector Road**: Typically contains lower traffic volume than either secondary main roads or main roads and permits traffic between main roads or secondary main roads and local roads. Example: Hollis Road.
- **Local Road**: Typically contains the lowest traffic volume of all roads and either permits traffic flow from any of the above roads to residential subdivisions or acts as a connector between subdivisions. Example: Broadmeadow Drive.

Pavement Defect Value: With the exception of newly paved roads, virtually all roads have some defects. The PMP evaluates a number of defect types and the prevalence of those defects. Roads with more defects and/or defects over a greater percentage of their length achieve a higher pavement defect value. The pavement defects we evaluated include:

- **Random Cracks**: Transverse cracks caused by seasonal elongation and contraction of pavement. These will occur within a few years of any new bituminous pavement.
- **Transverse/Longitudinal Cracks**: Similar to random cracks, these lengthwise and crosswise cracks are caused by seasonal elongation and contraction of pavement. These will occur within a few years of any new bituminous pavement.
- **Alligator Cracks**: Also called “spider webbing” and often found at and around manholes, catch basins, and/or driveways where they meet the road.
- **Pot Holes**: A crater like hole of varying size caused by freezing and thawing of the road surface particularly in areas of other defects and cracks.
- **Raveling**: Raveling occurs when sections of the top pavement become loose and separate from the binder course. It indicates deteriorating bond between pavement layers.

- **Rutting, Corrugations, or Pushing:** Longitudinal depressions and “wheel ruts” exist in travel lanes or the edge of the roadway. They are due primarily to repetitive pressure along one line of travel causing improper sub-base to compact and/or move laterally.
- **Cross Trenches, Utility Cuts, or other Patches:** These are sections of pavement that have been repaired with new bituminous pavement.
- **Main Line Trenches:** Main line trenches are longitudinal sections of pavement that have been repaired with new bituminous concrete pavement usually after a main line utility repair.

Survey Forms: The functional classification and pavement defect values were arrived at by using the Road Classification System Survey Form attached to this report. Each road or road segment was evaluated and a survey form filled out for each. The Pavement Defects section of the form shows the types of defects evaluated and the intensity of those defects. Column 1 indicates no defect. Moving left to right the columns indicate progressively higher defects, reflecting the percentage of the road that suffers from the particular defect. Column 6 represents the severest defect. The values of each defect, ranging from 1 to 6, are added together to give the Pavement Defect Value, which is listed at the bottom of the survey form. A low pavement defect value indicates a road in good condition while a high pavement defect value indicates a road in poor condition.

Priority Condition: Pavement defect value alone will not provide a useful priority list for road repair. Using both pavement defect value and road classification provides a more useful and informed outcome. It is necessary to give a weighted value to the road classification so that roads with a higher classification receive the attention they deserve. For the Pavement Management Plan, the following values were applied:

- Main Road: 0.75
- Secondary Main Road: 0.50
- Collector Road: 0.25
- Local Road: 0.10

Multiplying the pavement defect value by the road classification values provides the priority condition value.

$$\text{Pavement defect value} \times \text{functional classification value} = \text{priority condition value}$$

This provides a priority value for each road or segment. All of the roads/segments can then be sorted by the priority value to provide a priority list of roads.

Improvement Methods

Once Overall Priority Conditions for each roadway segment are completed specific improvement methods are identified. The Annual Improvement Plan will be designed taking into consideration available funds and the roadway network as a whole. The improvement program will address immediate needs and long term maintenance of the roadway network. Immediate needs are those in which roadway segments are at or near failure. Long term maintenance needs are those that slow or stop the deterioration of the roadway segment, thereby extending the life of the segment.

Improvements will be tailored to the specific condition of the roadway segment. Improvements may include crack sealing, patching, replacing broken berms, shoulder work, infrared repairs, trench repairs, milling, overlay, and total or partial reclamation or reconstruction. Other roadway features will also be evaluated and factored into improvement costs prior to a specific overall cost estimate. These may include curbing, manholes, and general drainage repairs including catch basins. The most typical pavement application allows for the use of existing pavement and sub base as the foundation for a leveling course which serves to reshape the roadway profile. Due to the rutting of the failed surface this course is difficult to measure. Once this phase of the reconstruction is complete a top course is installed consisting of two inches of bituminous concrete consisting of slightly smaller aggregate.

Preventive Maintenance

In addition to addressing roads in poor or failing condition, an emphasis must be placed on preventive maintenance. It is important to address pavement problems as they develop and not allow them to reach the point of pavement failure. For this reason, minor repair techniques will be utilized to extend the life of roadways. For example, crack sealing is a fairly inexpensive way to prevent water from infiltrating into pavement, which can quickly lead to pavement failure.

Reconstruction

For roadways that are beyond repair, reconstruction will be necessary. In general, reconstruction should only be considered for main roads. This is because of the volume of traffic these roadways carry. It is generally not cost effective to reconstruct secondary and local roadways that do not carry heavy volumes. These roads can usually be sufficiently repaired by less costly means.

Annual Improvement Program

Every year prior to the commencement of the construction season a list of roadway improvements will be published. This Annual Improvement Program will be provided to the Board of Selectmen and made available to the public on the Town's website. The program will detail the roadway segments and the corresponding pavement method for each segment. The decisions will be directly tied to the Pavement Management Program. Due to uncertain funding and expenses it is difficult to develop a list that projects when each roadway segment will be addressed. Multiyear improvement schedules will be released to the extent that they can be accurately projected.

Summary

A Pavement Management Program that incorporates roadway functional classification and pavement defect value provides an objective way of evaluating and prioritizing necessary roadway repairs. As noted above, roadway maintenance is funded from two sources: State c. 90 funding at approximately \$320,000 per year and local funds at about \$70,000 per year. That is not a lot of money to maintain an asset valued at approximately \$70,000,000. Absent an infusion of additional funds we must assume this is our annual allocation. Therefore, we must be as efficient as possible with our available funds. Adherence to the Pavement Management Program will give us the best ability to maintain this asset in as efficient a means as possible.

**ROAD CLASSIFICATION SYSTEM
SURVEY FORM**

Street Name _____
Section Beginning _____
Section Ending _____

Survey Date _____
Pavement Width _____ feet
Pavement Length _____ feet

Functional Class

- 1. Main Road ☐
- 2. Secondary Main ☐
- 3. Collector ☐
- 4. Local ☐

Other Features

- | | | |
|-----------------|------------------------------|-----------------------------|
| 1. Curbing/Berm | Yes ☐ | No ☐ |
| Utilities | ☐ | |
| Manholes | ☐ | |
| Catch Basins | ☐ | |

Proposed Repair (H, M, L, N)

- 1. Routine Maintenance
 - a. Crack filling ☐
 - b. Patching ☐
 - c. Replace broken berm ☐
 - d. Shoulder work ☐
 - e. Infra-red repairs ☐
 - f. Trench repairs ☐
- 2. Level & Overlay ☐
- 3. Complete Reconstruction
 - a. Total reclamation ☐
 - b. Partial reclamation ☐

Pavement Defects

	1	2	3	4	5	6
	0%	1-10%	11-25%	26-50%	51-75%	76-100%
1. Random Cracks	☐	☐	☐	☐	☐	☐
2. Transverse/Longitudinal Cracks	☐	☐	☐	☐	☐	☐
3. Alligator Cracks	☐	☐	☐	☐	☐	☐
4. Pot Holes	☐	☐	☐	☐	☐	☐
5. Raveling	☐	☐	☐	☐	☐	☐
6. Rutting, Corrugations, or Pushing	☐	☐	☐	☐	☐	☐
7. Cross Trenches, Utility Cuts, Other Patches	☐	☐	☐	☐	☐	☐
8. Main Line Trenches	☐	☐	☐	☐	☐	☐

Pavement Defect Value: _____

Comments:

Comprehensive Pavement Management Plan, by Priority Order

NAME	BEGINNING	ENDING	MILES	SURVEY DATE	PVMT WIDTH	PVMT LENGTH	PVMT DEF VAL	CURB/ BERM	FUNCTIONAL CLASS	FUNC CLASS VALUE	PRIORITY CONDITION	COMMENTS
NORTHFIELD ROAD 2	NW TOWNSEND ROAD	MAPLEWOOD GOLF COURSE	0	9/15/09 20	0.4	38			MAIN	0.75	28.5	
SUMMER STREET	FITCHBURG LINE	LEOMINSTER LINE	0.8	9/17/09 52.57	0.8	35	X		MAIN	0.75	26.25	Wide pavement, no markings = dangerous, full depth recommended
NW TOWNSEND RD 3	200M NO. 557 NW TOWNSEND	FITCHBURG LINE	0	9/15/09 22	1.8	33			MAIN	0.75	24.75	
NW TOWNSEND RD 2	FITCHBURG LINE @ HOWARD ST	200M NO. 557 NW TOWNSEND	0	9/15/09 22	0.5	31			MAIN	0.75	23.25	Significant deterioration
NORTHFIELD ROAD	NEW WEST TOWNSEND	FITCHBURG LINE	3.45	9/15/09 20	0.1	29			MAIN	0.75	21.75	
LEOM-SHIRLEY RD	LEOMINSTER LINE	PIONEER DRIVE	1.94	9/21/09 30	0.35	28			MAIN	0.75	21	L&O+T93 2010
NW TOWNSEND RD	TOWNSEND LINE	FITCHBURG LINE	1.34	9/15/09 22	0.5	28			MAIN	0.75	21	Potential TIP project w/Lun & Flth to reconstruct
RESERVOIR ROAD	LEOMINSTER-SHIRLEY RD	GOODRICH STREET	3.38	9/21/09 30	0.52	28			MAIN	0.75	21	
NORTHFIELD ROAD 5	100 M SE OF GILCHREST ST	OAK AVENUE	0	9/15/09 22-24	0.6	26			MAIN	0.75	19.5	Recommend overlay
FORT POND ROAD	LEOMINSTER LINE	LEOMINSTER-SHIRLEY RD	0	9/21/09 28	0.25	25			MAIN	0.75	18.75	OVERLAY WIN 5 YEARS
LAKEFRONT	PROSPECT STREET	LEOMINSTER LINE	0.29	9/18/09 32.52	0.29	36	X		SECONDARY	0.5	18	Total part. 52' - Includes parking - berm 1 side only
CHASE ROAD 2	C. MASS GARDEN CNTR	MASS AVENUE	0	9/15/09 36-42	1.8	23			MAIN	0.75	17.25	Overlay 3-5 yrs; small sec curbed near Mass ave
LANCASTER AVE 3	KILBURN STREET	GOODRICH STREET	0	9/18/09 22-24	0.6	23			MAIN	0.75	17.25	L&O win 5 years
LANCASTER AVE 4	GOODRICH STREET	GIBSON STREET	0	9/18/09 22	1.3	23			MAIN	0.75	17.25	L&O win 5 years
LANCASTER AVE 5	GIBSON STREET	LEOMINSTER-SHIRLEY RD	0	9/18/09 22	0.6	23			MAIN	0.75	17.25	
NORTHFIELD ROAD 3	MAPLEWOOD GOLF COURSE	CHASE ROAD	0	9/15/09 20	0.75	23			MAIN	0.75	17.25	
CHASE ROAD	TOWNSEND LINE	C.MASS GARDEN CNTR	2.79	9/15/09 36-40	0.95	22	X		MAIN	0.75	16.5	Overlay in 3-5 years, some curbing
NORTHFIELD ROAD 4	CHASE ROAD	100M SE OF GILCHREST ST	0	9/15/09 22-24	1.15	22			MAIN	0.75	16.5	Rec overlay; cb sunken in shoulder - hazard
GOODRICH ST 4	PAGE STREET	KILBURN AVENUE	0	9/18/09 24	0.65	21			MAIN	0.75	15.75	L&O win 5 years
WHITE STREET	MASS AVENUE	LANCASTER AVENUE	0	9/21/09 20	1.1	30			SECONDARY	0.5	15	
GOODRICH ST 2	RESERVOIR ROAD	FITCHBURG LINE	0.72	9/17/09 24	0.72	29			SECONDARY	0.5	14.5	Rocks pushing thru pavement; 2able base
WEST STREET 4	HOLLIS ROAD	721 GOODRICH STREET	0	9/21/09 18	0.2	25			SECONDARY	0.5	12.5	L&O 2010
LEOMINSTER ROAD 2	PROSPECT STREET	SUNNY HILL ROAD	0	9/18/09 19-24	0	25			SECONDARY	0.5	12.5	
RESERVOIR RD 3	FLAT HILL ROAD	200 M NORTH OF KILBURN ST	0	9/18/09 31	0.7	16			MAIN	0.75	12	
RESERVOIR RD 2	GOODRICH STREET	297 RESERVOIR ROAD	0	9/21/09 24	0.3	16			MAIN	0.75	12	
RESERVOIR RD 4	297 RESERVOIR	FLAT HILL ROAD	0	9/21/09 30	1.4	15			MAIN	0.75	11.25	Recommend overlay near sand pit
NORTHFIELD ROAD 6	OAK AVENUE	PAGE STREET	0	9/21/09 24	0.91	15			MAIN	0.75	11.25	
GOODRICH ST 3	721 GOODRICH STREET	MASS AVENUE	0	9/15/09 22-24	0.35	14			MAIN	0.75	10.5	
W TOWNSEND RD 2	CHASE ROAD	587 GOODRICH STREET	0	9/21/09 18	0.4	19			SECONDARY	0.5	9.5	
FLAT HILL ROAD 4	RESERVOIR ROAD	262 WEST TOWNSEND ROAD	0	9/15/09 24-26	0.3	19			SECONDARY	0.5	9.5	
HOLMAN STREET	CHASE ROAD	SUNSET LAND	0	9/21/09 18	0.9	18			SECONDARY	0.5	9	
HOLMAN STREET 3	NORTHFIELD ROAD	1 LANE BRIDGE (MULPLUS BRK)	1.7	9/15/09 18-20	0.45	18			SECONDARY	0.5	9	
W TOWNSEND RD 3	262 WEST TOWNSEND ROAD	HIGHLAND STREET	0	9/15/09 20-22	0.55	18			SECONDARY	0.5	9	
HOWARD STREET	NEW WEST TOWNSEND-HH59 ROAD	TOWNSEND LINE	0	9/15/09 24-26	1.1	18			SECONDARY	0.5	9	
FLAT HILL ROAD 2	COURTLAND CIRCLE SOUTH	55 HOWARD STREET	1.78	9/15/09 26	0.2	36			COLLECTOR	0.25	9	
LEOM-SHIRLEY RD 3	FORT POND ROAD	ELMWOOD ROAD	0	9/25/09 22	0.51	17			SECONDARY	0.5	8.5	
FLAT HILL ROAD	PAGE STREET	SHIRLEY LINE	0	9/21/09 28	1.1	11			MAIN	0.75	8.25	
ELECTRIC AVENUE	LEOMINSTER LINE	COURTLAND CR (SOUTH)	1.26	9/25/09 19	0.3	16			SECONDARY	0.5	8	
LANCASTER AVE	MASS AVENUE	MASSACHUSETTS AVE	1.58	9/18/09 32-40	1.58	10	X		MAIN	0.75	7.5	
LEOMINSTER ROAD	LANCASTER AVENUE	PAGE STREET	4.36	9/18/09 28	1.1	10	X		MAIN	0.75	7.5	
TOWNSEND HBR RD	TOWNSEND LINE	PROSPECT STREET	2.55	9/18/09 26	1.1	10	X		MAIN	0.75	7.5	
FLAT HILL ROAD 3	SUNSET LANE	MASS AVENUE	1.87	9/21/09 25	1.81	10			MAIN	0.75	7.5	
WEST STREET 2	727 WEST STREET	PAGE ROAD	0	9/21/09 18	0.85	15			SECONDARY	0.5	7.5	
WHALOW ROAD 2	SUMMER STREET	WEST STREET EXT	0	9/17/09 20	0	15			SECONDARY	0.5	7.5	
KILBURN STREET 2	GOODRICH STREET	ELECTRIC AVENUE	0	9/17/09 26-30	0	15	X		SECONDARY	0.5	7.5	
WEST STREET	ELECTRIC AVENUE	LEOMINSTER ROAD	0	9/18/09 22	0.3	30			COLLECTOR	0.25	7.5	
LEOMINSTER ROAD 3	200 M NORTH OF KILBURN ST	727 WEST STREET	2.45	9/17/09 20-24		14			SECONDARY	0.5	7	
SUNNY HILL ROAD	WEST STREET	LEOMINSTER LINE	0	9/18/09 29	0.7	9			MAIN	0.75	6.75	
PAGE STREET 4	FLAT HILL ROAD	MASS AVENUE	1.43	9/18/09 19		13			SECONDARY	0.5	6.5	
LEOM-SHIRLEY RD 2	PIONEER DRIVE	416 PAGE STREET	0	9/25/09 18	0.51	25			COLLECTOR	0.25	6.25	
		FORT POND ROAD	0	9/21/09 33	0.4	8			MAIN	0.75	6	

Comprehensive Pavement Management Plan, by Priority Order

NAME	BEGINNING	ENDING	MILES	SURVEY DATE	PVMT WIDTH	PVMT LENGTH	PVMT DEF VAL	CURB/ BERM	FUNCTIONAL CLASS	FUNC CLASS VALUE	PRIORITY CONDITION	COMMENTS
PAGE STREET 2	LANCASTER AVENUE	387 PAGE STREET	0	9/25/09 21	1.05	24			COLLECTOR	0.25	6	
YOUNGS ROAD	WHALOM ROAD	SUMMER STREET	0.54	9/17/09 20	0.54	23			COLLECTOR	0.25	5.75	
W TOWNSEND RD	CHASE ROAD	NORTHFIELD ROAD	1.51	9/15/09 28	0.3	11			SECONDARY	0.5	5.5	
GOODRICH STREET	LANCASTER AVENUE	KILBURN STREET	2.42	9/18/09 23	0.57	10			SECONDARY	0.5	5	
PIONEER DRIVE	LEOMINSTER-SHIRLEY ROAD	LEOMINSTER LINE	0.2	9/21/09 39	0.2	10	X		SECONDARY	0.5	5	
SUNNY HILL ROAD 2	WEST STREET	LEOMINSTER ROAD	0	9/18/09 18	0	10			SECONDARY	0.5	5	
WHALOM ROAD	ELECTRIC ROAD	LAKEFRONT ROAD	1.27	9/18/09 26-32		10			SECONDARY	0.5	5	
ELMWOOD ROAD	MASS AVENUE	END	1.42	11/1/09 20	0.95	20			COLLECTOR	0.25	5	Strategic L&O sub considered for worst segmt
TURKEY HILL ROAD	NORTHFIELD ROAD	OAK AVENUE	0.38	9/15/09 20-22	0.38	19			COLLECTOR	0.25	4.75	Pavement sags inward from shoulders
HOLMAN STREET 2	ONE LANE BRIDGE	NORTHFIELD ROAD	0	9/15/09 22	0.5	9			SECONDARY	0.5	4.5	
MAIN STREET	MASS AVENUE	OAK AVENUE	0.31	9/15/09 24	0.31	9	X		SECONDARY	0.5	4.5	
MULPUS ROAD	TOWNSEND HBR ROAD	SHIRLEY LINE	1.83	9/21/09 20	1.83	9			SECONDARY	0.5	4.5	
ARBOR STREET	PAGE STREET	MASS AVE	1.26	11/1/09 22	1.25	18			COLLECTOR	0.25	4.5	Overlay win 5 yrs
BEAL STREET	CHASE ROAD	MASS AVE	0.5	9/15/09 18-88	0.5	18			COLLECTOR	0.25	4.5	Overlay is ideal treatment; patch/seal yrs 1-5
BURRAGE STREET	PAGE STREET	FLAT HILL ROAD	1.83	9/21/09 20	1.25	18			COLLECTOR	0.25	4.5	
HOLLIS ROAD 2	WEST STREET	PROSPECT STREET	0	9/18/09 20-22	0.7	18			COLLECTOR	0.25	4.5	
KILBURN STREET	LANCASTER AVENUE	GOODRICH STREET	0.98	9/18/09 22	0.6	18			COLLECTOR	0.25	4.5	
HOWARD STREET 2	55 HOWARD STREET	MATCH LINE 14/1277	0	9/15/09 26	0.15	17			COLLECTOR	0.25	4.25	
WEST STREET 3	ELECTRIC AVENUE	HOLLIS ROAD	0	9/18/09 19	0	8			SECONDARY	0.5	4	Repaved 2010 - segment in perfect condition
WEST STREET 5	SUNNY HILL ROAD	LEOMINSTER ROAD	0	9/18/09 18-20	0	8			SECONDARY	0.5	4	
BURRAGE STREET 2	FLAT HILL ROAD	SHIRLEY LINE	0	9/21/09 23	0.5	16			COLLECTOR	0.25	4	
HOLLIS ROAD	MASS AVENUE	WEST STREET	1.43	9/18/09 20-22	0.7	16			COLLECTOR	0.25	4	
CHESTNUT STREET	MASS AVENUE	HIGHLAND STREET	0.47	9/15/09 18	0.47	15			COLLECTOR	0.25	3.75	
HOWARD STREET 3	14/1277 MATCH LINE	NW TOWNSEND ROAD	0	9/15/09 26-30	1.35	15			COLLECTOR	0.25	3.75	
PLEASANT STREET	WEST STREET	MASS AVENUE	0.96	9/17/09 24.26	0.96	15			COLLECTOR	0.25	3.75	
PRATT STREET	WEST STREET	WHALOM ROAD	0.38	9/17/09 20	0.38	15			COLLECTOR	0.25	3.75	
WILDERWOOD AVE	LEOMINSTER LINE	LEOMINSTER LINE	0.15	4/14/10 21		37			LOCAL	0.1	3.7	
WATT STREET	FLAT HILL ROAD	PAGE STREET	0.3	9/25/09 14	0.3	35			LOCAL	0.1	3.5	Full depth recommended
MAPLE PARKWAY	WHITE STREET	END	0.26	9/17/09 20	0.26	34			LOCAL	0.1	3.4	L&O ideal repair, justifiable; CB's settling, 7able base
PLEASANTVIEW AVE	GRAHAM STREET	END	0.1	9/17/09 18.22	0.1	33			LOCAL	0.1	3.3	Significant deterioration; full depth recommended
PAGE HILL ROAD	PEARL BROOK ROAD	BROADMEADOW DRIVE	0.19	9/17/09 29	0.19	32	X		LOCAL	0.1	3.2	Partial berm
PEARL BROOK RD	WHITE STREET	BROADMEADOW DRIVE	0.17	9/17/09 29	0.17	32			LOCAL	0.1	3.2	
GIBSON STREET	LANCASTER AVENUE	LEOMINSTER LINE	0.26	9/18/09 20-22	0.26	31			LOCAL	0.1	3.1	
UPLAND AVENUE	HOLLIS ROAD	ELECTRIC AVENUE	0.26	9/18/09 22	0.26	12	X		COLLECTOR	0.25	3	
LAKESIDE AVENUE	LEOMINSTER LINE	WILDERWOOD ROAD	0.1	4/14/10 21		29			LOCAL	0.1	2.9	
WILDSIDE AVENUE	BEACHVIEW ROAD	SOUTH ROW ROAD	0.34	9/21/09 20	0.34	29			LOCAL	0.1	2.9	
KIMBALL STREET	PROSPECT STREET	POND STREET	0.13	9/18/09 25	0.13	28			LOCAL	0.1	2.8	
HIGHLAND STREET	MAIN STREET	NORTHFIELD ROAD	1.19	9/15/09 24	1.19	11	X		COLLECTOR	0.25	2.75	Crack sealing
CRESCENT TER	CRESCENT ROAD	END	0.07	9/21/09	0.07	26			LOCAL	0.1	2.6	
EASTERN AVENUE	ELECTRIC AVENUE	HOLLIS ROAD	0.26	9/18/09 30	0.26	10			COLLECTOR	0.25	2.5	
PAGE STREET 3	387 PAGE STREET	416 PAGE STREET	0	9/25/09 21	0.12	10			COLLECTOR	0.25	2.5	
PINE STREET	ARBOR STREET	MASS AVENUE	0.25	11/1/09 21	0.21	10			COLLECTOR	0.25	2.5	
PROSPECT STREET	LEOMINSTER ROAD	WHALOM ROAD	1.04	9/18/09 26	1.04	10			COLLECTOR	0.25	2.5	
ROUND ROAD	RESERVOIR ROAD	END	0.79	9/21/09 18	0.79	25			LOCAL	0.1	2.5	
SOUTH ROW ROAD	TOWNSEND HARBOR ROAD	TOWNSEND LINE	0.72	9/21/09 23	0.72	25			LOCAL	0.1	2.5	
WEST GROTON RD	OLD MASS AVENUE	SHIRLEY LINE	0.13	9/25/09 24	0.13	25			LOCAL	0.1	2.5	
BEACHVIEW ROAD	ISLAND ROAD	WILDWOOD ROAD	0.27	9/21/09 20	0.27	24			LOCAL	0.1	2.4	
FISH STREET	LEOMINSTER ROAD	LEOMINSTER LINE	0.29	9/18/09 20	0.29	24			LOCAL	0.1	2.4	
RENNIE STREET	PRATT STREET	END	0.13	9/17/09 20	0.13	24			LOCAL	0.1	2.4	Significant deterioration; low traffic
CRESCENT ROAD	TOWNSEND HARBOR RD	SOUTH ROW ROAD	0.12	9/21/09 21	0.12	23			LOCAL	0.1	2.3	
PAGE STREET	FLAT HILL ROAD	SHIRLEY LINE	2.39	9/25/09 18	0.52	9			COLLECTOR	0.25	2.25	

Comprehensive Pavement Management Plan, by Priority Order

NAME	BEGINNING	ENDING	MILES	SURVEY DATE	PVMT WIDTH	PVMT LENGTH	PVMT DEF VAL	CURB/ BERM	FUNCTIONAL CLASS	FUNG CLASS VALUE	PRIORITY CONDITION	COMMENTS
CROSS ROAD	FISH STREET	PROSPECT STREET	0.95	9/18/09 20		0.92	22		LOCAL	0.1	2.2	
CROSS STREET	MASS AVENUE	MULPUS ROAD	0.42	9/25/09 22		0.41	22		LOCAL	0.1	2.2	
ISLAND ROAD	SOUTH ROW ROAD	END	0.58	9/21/09 20		0.58	22		LOCAL	0.1	2.2	
PIEDMONT AVENUE	SUMMER STREET	END	0.12	9/17/09 22		0.12	22		LOCAL	0.1	2.2	Repair sewer trench
RAMGREN ROAD	WHALOM ROAD	END	0.2	9/17/09 20		0.2	22		LOCAL	0.1	2.2	
SANDY COVE ROAD	SOUTH ROW ROAD	COVE ROAD	0.35	9/21/09 22		0.35	21		LOCAL	0.1	2.1	
WILLIAMS DRIVE	ISLAND ROAD	END		9/21/09 24			21	X	LOCAL	0.1	2.1	
OAK AVENUE	MAIN STREET	NORTHFIELD ROAD	0.45	9/15/09 28.3		0.45	8	X	COLLECTOR	0.25	2	
WHITING STREET	LEOMINSTER ROAD	MASS AVENUE	0.42	11/1/09 19		0.32	8		COLLECTOR	0.25	2	
COVE ROAD	SOUTH ROW ROAD	SANDY COVE ROAD	0.43	9/21/09 22		0.43	20		LOCAL	0.1	2	
PENINSULA DRIVE	SOUTH ROW ROAD	END	0.55	9/21/09 20		0.55	19		LOCAL	0.1	1.9	
SAVAGE DRIVE	LEOMINSTER ROAD	END	0.06	4/14/10	15		19		LOCAL	0.1	1.9	
AUTUMN ROAD	RESERVOIR ROAD	END	0.29	9/21/09 20		0.29	18	X	LOCAL	0.1	1.8	
HORIZON ISLAND RD	ISLAND ROAD	END	0.28	9/21/09 20		0.28	18		LOCAL	0.1	1.8	
HUNTING HILL ROAD	MULPUS ROAD	END	0.33	9/21/09 16		0.33	18		LOCAL	0.1	1.8	
CHARLTON STREET	WHITE STREET	FITCHBURG LINE	0.22	9/17/09 28		0.22	17		LOCAL	0.1	1.7	
SCHOOL STREET	MASS AVENUE	END	0.08	11/1/09			17		LOCAL	0.1	1.7	
WILDER ROAD	LEOMINSTER LINE	CROCKER AVENUE		4/14/10	20		17		LOCAL	0.1	1.7	
WINDWARD TER	BEAL STREET	END	0.11	9/15/09 30		0.4	17	X	LOCAL	0.1	1.7	Crack seal/patch
BOUCHER ROAD	CHASE ROAD	END	0.3	9/15/09 20-22		0.3	16		LOCAL	0.1	1.6	
COUNTRY ROAD	CHASE ROAD	END	0.14	9/15/09 26		0.14	16		LOCAL	0.1	1.6	CB's & M/H settlement
CREST AVENUE	HOLLIS ROAD	END	0.13	9/18/09 20		0.13	16		LOCAL	0.1	1.6	
GILCHREST STREET	NORTHFIELD ROAD	VALLEY ROAD	0.56	9/15/09 22-24		0.56	16		LOCAL	0.1	1.6	Minor rutting, 1st 100 m badly deteriorated
LAKEVIEW AVENUE	CROSS STREET	SUNSET LANE		9/18/09 12-20			16		LOCAL	0.1	1.6	
CANTERBURY DRIVE	ARBOR STREET	END		11/1/09 20			15	X	LOCAL	0.1	1.5	Settling/collapsing @ catch basins
CUSHING LANE	TURKEY HILL ROAD	END	0.17	9/15/09 20-22		0.17	15		LOCAL	0.1	1.5	
THE LANE	ELMWOOD ROAD	MASS AVENUE		9/25/09 17		0.31	15		LOCAL	0.1	1.5	
COVE TERRACE	COVE ROAD	END	0.06	9/21/09 20		0.06	14		LOCAL	0.1	1.4	
HOUGHTON MILL RD	RESERVOIR ROAD	END		9/21/09 24			14	X	LOCAL	0.1	1.4	
STONE FENCE ROAD	FALT HILL ROAD	END		9/21/09 22			14	X	LOCAL	0.1	1.4	
VALLEY ROAD	CLIFFVIEW ROAD	GILCHRIST STREET	0.33	4/14/10	22		14		LOCAL	0.1	1.4	
LONGWOOD DRIVE	BURRAGE STREET	BURRAGE STREET		9/21/09 24			13	X	LOCAL	0.1	1.3	
CONNELL DRIVE	MASS AVENUE	END		11/1/09 21			12	X	LOCAL	0.1	1.2	
MEMORIAL DRIVE	MAIN STREET	MASS AVENUE	0.16	9/18/09 24-30		0.16	12		LOCAL	0.1	1.2	
PINE GROVE ROAD	LAKEVIEW AVENUE	CROSS ROAD	0.24	9/18/09 22		0.24	12	X	LOCAL	0.1	1.2	
ROBBS HILL ROAD	SUNSET LANE	SHIRLEY LINE		4/14/10	25		12		LOCAL	0.1	1.2	
RUTH STREET	PARMENTER ROAD	END		9/21/09 24			12		LOCAL	0.1	1.2	
TILTON AVENUE	MASS AVENUE	END	0.14	9/17/09 20		0.14	12		LOCAL	0.1	1.2	
WEST STREET TER	WEST STREET	WEST STREET	0.27	9/17/09 20		0.27	12		LOCAL	0.1	1.2	
BURKE STREET	NATICK STREET	JOSLIN STREET	0.14	9/17/09 20		0.14	11	X	LOCAL	0.1	1.1	
CROCKER AVENUE	LEOMINSTER LINE	END	0.1	4/14/10	16		11		LOCAL	0.1	1.1	
GRAHAM STREET	SUMMER STREET	WHALOM ROAD	0.28	9/17/09 23		0.28	11		LOCAL	0.1	1.1	
HILLSIDE DRIVE	MASS AVENUE	END	0.1	11/1/09 20			11		LOCAL	0.1	1.1	
JOSLIN STREET	BURKE STREET	WHALOM ROAD	0.15	9/17/09 20		0.15	11	X	LOCAL	0.1	1.1	
MAYS FIELD ROAD	RICHARDS WAY	END		9/17/09 24			11	X	LOCAL	0.1	1.1	
POND STREET	PROSPECT STREET	HOLLIS ROAD	0.2	9/18/09 24		0.2	11		LOCAL	0.1	1.1	
SUNSET LANE	FLAT HILL ROAD	END		4/14/10	24		11	X	LOCAL	0.1	1.1	50% berm
BROWN AVENUE	PLEASANT STREET	END	0.06	9/17/09 8-16		0.06	10		LOCAL	0.1	1	
CLIFFVIEW TER	VALLEY ROAD	END	0.16	9/15/09 18.20		0.16	10		LOCAL	0.1	1	
CLIFTON ROAD	WEST STREET	RANGELEY ROAD	0.12	9/18/09 28		0.12	10		LOCAL	0.1	1	
EAST STREET	YOUNGS ROAD	END	0.14	9/17/09 20		0.14	10		LOCAL	0.1	1	

Comprehensive Pavement Management Plan, by Priority Order

NAME	BEGINNING	ENDING	MILES	SURVEY DATE	PVMT WIDTH	PVMT LENGTH	PVMT DEF VAL	CURB/BERM	FUNCTIONAL CLASS	FUNC CLASS VALUE	PRIORITY CONDITION	COMMENTS
EDITH LANE	RICHARDS WAY	END		9/17/09 24			10	X	LOCAL	0.1	1	
ELM STREET	PROSPECT STREET	SUNNYHILL ROAD	0.45	9/18/09 24		0.45	10		LOCAL	0.1	1	
FAIRVIEW ROAD	LEOMINSTER ROAD	END	0.33	11/1/09 20			10		LOCAL	0.1	1	
HEMLOCK DRIVE	TOWNSEND HARBOR ROAD	END	0.53	9/21/09 19		0.53	10		LOCAL	0.1	1	
LAUREL LANE	PROSPECT STREET	END	0.27	9/18/09 18		0.27	10		LOCAL	0.1	1	
LESURE AVENUE	PIERCE AVENUE	WHALOM ROAD	0.2	9/17/09 21		0.2	10		LOCAL	0.1	1	
LINCOLN STREET	LEOMINSTER LINE	FISH STREET	0.19	9/18/09 16		0.19	10		LOCAL	0.1	1	
MEADOW LANE	ROLLING ACRES ROAD	ROLLING ACRES ROAD	0.15	11/1/09 22			10		LOCAL	0.1	1	
PARMENTER ROAD	ROUND ROAD	RUTH STREET		9/21/09 24			10		LOCAL	0.1	1	
PIERCE AVENUE	LEOMINSTER LINE	ELECTRIC AVENUE	0.46	9/17/09 21		0.46	10		LOCAL	0.1	1	
RANGELY ROAD	CLIFTON ROAD	REDWOOD ROAD	0.15	9/18/09 28		0.15	10		LOCAL	0.1	1	
REDWOOD ROAD	RANGELEY ROAD	WEST STREET	0.08	9/18/09 28		0.08	10		LOCAL	0.1	1	
RICHARDS WAY	MASS AVENUE	END		9/17/09 24			10	X	LOCAL	0.1	1	Some settling of structures
ROLLING ACRES RD	LANCASTER AVENUE	MEADOW LANE	0.7	11/1/09 22			10		LOCAL	0.1	1	
SKYLARK LANE	BEAL STREET	END	0.13	9/15/09 22		0.13	10		LOCAL	0.1	1	Crack sealing years 3-5
WEST ACRES DRIVE	WEST STREET TERRACE	WEST STREET TERRACE	0.2	9/17/09 20-24		0.2	10		LOCAL	0.1	1	
BAKER STREET	WEATHERBEE STREET	KINGMAN STREET	0.1	9/17/09 20-22		0.1	9		LOCAL	0.1	0.9	
BROOKVIEW TER	VALLEY ROAD	END	0.17	9/15/09 22-24		0.17	9	X	LOCAL	0.1	0.9	
CORTLAND CIRCLE	FLAT HILL ROAD	FLAT HILL ROAD		9/25/09 22		0.11	9		LOCAL	0.1	0.9	
ELIZABETH STREET	PIERCE AVENUE	WHALOM ROAD	0.11	9/17/09 24		0.11	9		LOCAL	0.1	0.9	Crack sealing only, circle in poor condition
FITCHVIEW AVENUE	ELECTRIC AVENUE	END	0.09	9/17/09		0.09	9		LOCAL	0.1	0.9	
HAMLIN STREET	BURKE STREET	WHALOM ROAD	0.16	9/17/08 20		0.16	9	X	LOCAL	0.1	0.9	
JOHN STREET	WHALOM ROAD	END	0.17	9/17/09 19		0.17	9		LOCAL	0.1	0.9	
KINGMAN STREET	SUMMER STREET	YOUNGS ROAD	0.12	9/17/09 20		0.12	9		LOCAL	0.1	0.9	
NATICK STREET	WHALOM ROAD	BURKE STREET	0.14	9/17/09 20		0.14	9	X	LOCAL	0.1	0.9	
OTIS STREET	WHALOM ROAD	BURKE STREET	0.15	9/17/09 20		0.15	9	X	LOCAL	0.1	0.9	
SHAKER COURT	BURRAGE STREET	END		4/14/10 24			9	X	LOCAL	0.1	0.9	
THE CLEARING	WHALOM ROAD	PIERCE AVENUE	0.15	9/17/09 14,18			9		LOCAL	0.1	0.9	
WEST STREET EXT	WEST ST @ WEST ST TERR	END	0	9/17/09 16-18		0	9		LOCAL	0.1	0.9	
WOODLAND DRIVE	TOWNSEND HARBOR ROAD	TOWNSEND HARBOR ROAD	0.47	9/21/09 20		0.47	9		LOCAL	0.1	0.9	
ALDER WAY	COURTLAND CIRCLE	End		9/25/09 22			8	X	LOCAL	0.1	0.8	
ANDREW TERRACE	FITCHVIEW AVENUE	End	0.1	9/17/09 22		0.1	8		LOCAL	0.1	0.8	
BIRCH ISLAND WAY	HEMLOCK DRIVE	END	0.15	9/21/09 18		0.15	8	X	LOCAL	0.1	0.8	
DANA STREET	WHALOM ROAD	END		9/17/09			8	X	LOCAL	0.1	0.8	Gravel road
EASTER BROOK RD	GOODRICH STREET	END		9/21/09 24			8	X	LOCAL	0.1	0.8	
FLORENCE STREET	PIERCE AVENUE	WHALOM ROAD	0.12	9/17/09 22		0.12	8		LOCAL	0.1	0.8	
FRANCIS AVENUE	STEVENS STREET	END		11/1/09			8		LOCAL	0.1	0.8	
GABES PLACE	RESERVOIR ROAD	END		9/21/09 18			8	X	LOCAL	0.1	0.8	
HICKORY LANE	PINE ACRES ROAD	END	0.04	9/21/09 18		0.04	8		LOCAL	0.1	0.8	
JACKLYN DRIVE	MAY'S FIELD DRIVE	END	0	9/17/09 24		0	8	X	LOCAL	0.1	0.8	
JOHNSON STREET	PARMENTER/RUTH	END		9/21/09 24			8		LOCAL	0.1	0.8	
KIRBY AVENUE	WALLIS PARK	ELECTRIC AVENUE	0.12	9/17/09 22			8		LOCAL	0.1	0.8	
KIRBY AVENUE 2	FLORENCE STREET	END	0	9/17/09 22		0	8		LOCAL	0.1	0.8	
OLD FARM ROAD	NORTHFIELD ROAD	END		9/15/09 24			8	X	LOCAL	0.1	0.8	
PINE ACRES ROAD	HEMLOCK DRIVE	END	0.21	9/21/09 18		0.21	8		LOCAL	0.1	0.8	
ROBBS TERRACE	ROBB'S HILL ROAD	OAK RIDGE ROAD		4/14/10 25			8		LOCAL	0.1	0.8	
SEQUOIA DRIVE	HOWARD STREET	END		4/14/10 23			8		LOCAL	0.1	0.8	
STEVENS STREET	LEOMINSTER ROAD	FRANCIS AVENUE	0.06	11/1/09 18			8		LOCAL	0.1	0.8	
TOWN STREET	FISH STREET	END		9/18/09 16			8		LOCAL	0.1	0.8	
WALLIS PARK	WHALOM ROAD	PIERCE AVENUE	0.13	9/17/09 22		0.13	8		LOCAL	0.1	0.8	
WEATHERBEE ST	YOUNGS ROAD	SUMMER STREET	0.11	9/17/09 20-22		0.11	8		LOCAL	0.1	0.8	

Comprehensive Pavement Management Plan, by Priority Order

NAME	BEGINNING	ENDING	MILES	SURVEY DATE	PVMT WIDTH	PVMT LENGTH	PVMT DEF VAL	CURB/ BERM	FUNCTIONAL CLASS	FUNC CLASS VALUE	PRIORITY CONDITION	COMMENTS
WILLOW CIRCLE	COURTLAND CIRCLE	END		9/25/09	22		8	X	LOCAL	0.1	0.8	
WINDERMERE DRIVE	CHASE ROAD	END		9/17/09	24		8		LOCAL	0.1	0.8	Some settling @ structures
WINTER HILL ROAD	GOODRICH STREET	END		9/21/09	18		8	X	LOCAL	0.1	0.8	
ASPLUND DRIVE	BOUCHER ROAD	END		9/15/09			0				0	Gravel road
BOUTWELL STREET	YOUNGS ROAD	END		9/17/09			0		LOCAL	0.1	0	Gravel surface
BROADMEADOW DR	WHITE STREET	PAGE HILL ROAD	0.31	9/17/09	29	0.31	0	X	LOCAL	0.1	0	
BUTTONWOOD PL	WEST STREET	ELECTRIC AVENUE	0.23				0				0	
CARR AVENUE	WHALOM ROAD	END	0.33	9/18/09	18	0.33	0		LOCAL	0.1	0	Unpaved
FIRE ROAD 10							0				0	
FIRE ROAD 12							0				0	
FIRE ROAD 15	PARMENTER/ROUND	END		9/21/09	15		0		LOCAL	0.1	0	
FIRE ROAD 15A	FIRE RD 15	END		9/21/09			0				0	Gravel
FIRE ROAD 16							0				0	
FIRE ROAD 19							0				0	
FIRE ROAD 24B							0				0	
FIRE ROAD 7							0				0	
FLYNN ROAD							0				0	
HARBOR TERRACE	TOWNSEND HARBOR ROAD			9/21/09			0		LOCAL	0.1	0	
HARRIS AVENUE							0				0	
HEMLOCK TERRACE	HEMLOCK TERRACE	END	0.14	9/21/09		0.14	0		LOCAL	0.1	0	
HILL TOP LANE	HEMLOCK DRIVE	END	0.1	9/21/09		0.01	0		LOCAL	0.1	0	
IRIS COURT							0				0	
LAWTON AVENUE	FITCHBURG LINE	END	0.07	9/17/09	14-18	0.07	0		LOCAL	0.1	0	
MASS AVE							0		STATE		0	
MAY STREET							0				0	
OAK RIDGE ROAD							0				0	
PARK STREET TER	SOUTH ROW ROAD	END	0.06	9/21/09		0.06	0				0	Gravel
PEARL STREET							0				0	
RILEY ROAD							0				0	
ROYAL FERN DRIVE							0				0	
SEAVER ROAD							0				0	
SPRING STREET							0				0	
SPRING STREET EXT							0				0	
THORNDIKE STREET							0				0	
TOWNSEND STREET							0				0	
TRILLIUM COURT							0				0	
WINTERGREEN CT							0				0	
WOODBURY STREET							0				0	
FERDINAND PSWY							4				0	

LUNENBURG DEPARTMENT OF PUBLIC WORKS



PAVEMENT SURVEY AND MAINTENANCE PROGRAM

Highway Classification System (Functional Classes)

1. **Main Road** – Typically contains high traffic volume, and either permits traffic flow from one main road to another, or acts as an inter-town connector between the two. **Example: Lancaster Avenue**
2. **Secondary Main Road** – Typically contains higher traffic volume than either a collector road or a local road with less volume than a main road, also acting as a connector between main roads. **Example: Sunnyhill Road**
3. **Collector Road** – Typically contains lower traffic volume than either secondary main roads or main roads and permits traffic flow between main roads or secondary roads and local roads. **Example: Hollis Road**
4. **Local Road Over 1000 ft.** – Typically contains the lowest traffic volume of all roads and either permits traffic flow from any of the above roads to residential sub-divisions or acts as a connector between sub-divisions. **Example: Broadmeadow Drive**
5. **Local Road Under 1000 ft** – **Example: Windermere Drive**

ROAD CLASSIFICATION

#1 Lunenburg's Main Roads

Chase Road/Rt 13/Electric Avenue

Lancaster Avenue

Leominster Road

Leominster/Shirley Road

Townsend Harbor Road

New West Townsend Road

Summer Street

Northfield Road

Reservoir Road

ROAD CLASSIFICATION

#2 Lunenburg's Secondary

Main Roads

West Townsend Road
Holman Street
Flat Hill Road
Goodrich Street
Lakefront
Mulpus Road
Main Street
Pioneer Drive
Sunnyhill Road
West Street
Whalom Road
White Street

ROAD CLASSIFICATION

#3 Lunenburg's Collector Roads

Arbor Street
Beal Street
Burrage Street
Chestnut Street
Eastern Avenue
Elmwood Avenue
Hollis Road
Highland Street
Howard Street
Kilburn Street
Oak Avenue

Page Street
Pine Street
Pleasant Street
Pratt Street
Prospect Street
School Street
Sunset Lane
Turkey Hill Road
Upland Avenue
Whiting Street
Young's Road

ROAD CLASSIFICATION

#4 Lunenburg's Local Roads

(over 1000 ft. +/-) - Page 1

Broadmeadow Drive
Canterbury Drive
Connel Drive
Cove Road
Charlton Street
Easterbrook Road
Fairview Road
Fish Street
Bigson Street
Gilchrest Street
Hemlock Drive
Horizon Island Road
Houghton Mill Road

Island road
Laurel Lane
Longwood Drive
Maple Parkway
May's Field Road
Old Farm road
Penninsula Drive
Pierce Avenue
Pine Acres Road
Pine Grove Road
Richards Way
Robbs Hill Road
Rolling Acres Road

ROAD CLASSIFICATION

#4 Lunenburg's Local Roads

(over 1000 ft. +/-) - Page 2

Round Road	Watt Street
Ruth Street	West Acres Drive
Sandy Cove Road	West Street Terrace
Sequoia Drive	Wildwood Road
Shaker Court	Williams Road
South Row Road	Winterhill Road
Stone Fence Road	Woodland Drive
Valley Road	

ROAD CLASSIFICATION

#4 Lunenburg's Local Roads

(Under 1000 ft. +/-) Page 1

Andrew Terrace
Autumn Drive
Baker Street
Beachview Drive
Birth Island Way
Boucher Road
Brookview Terrace
Brown Avenue
Burke Street
Cliffview Terrace
Clifton Road
Country Road

Cove Terrace
Crescent Road
Crescent Terrace
Crest Avenue
Crocker Avenue
Cushing Lane
East Street
Elizabeth Street
Fitchview Avenue
Florence Street
Francis Avenue
Gabes Place

ROAD CLASSIFICATION

#4 Lunenburg's Local Roads

(Under 1000 ft. +/-) Page 2

Hamlin Street
Hickory Lane
Hillside Drive
John Street
Joslin Street
Kimball Street
Kingman Street
Kirby Avenue
Lakeview Avenue
Lawton Avenue
Lesure Avenue
Lincoln Street

Meadow Lane
Natick Street
Otis Street
Page Hill Road
Parmenter Road
Pearl Brook Road
Piedmont Avenue
Pleasantview Avenue
Pond Street
Ramgren Road
Rabgekt riad
Redwood Road

ROAD CLASSIFICATION

#4 Lunenburg's Local Roads

(Under 1000 ft. +/-) Page 3

Rennie Street
Robbs Terrace
Savage Drive
Skylark Lane
The Clearing
Tilton Avenue
Town Street
Wallis Park
Weatherbee Street
Wilder Road
Wilderwood Avenue
Windermere Drive
Windward Terrace

What type of road defects will be looked at in the survey...

Random/Longitudinal

Cracks

Utility Cuts/Other Patches

**Raveling, Shoving,
Pushing**

Main Line Trenches

Pot Holes

Rutting

Alligator Cracks



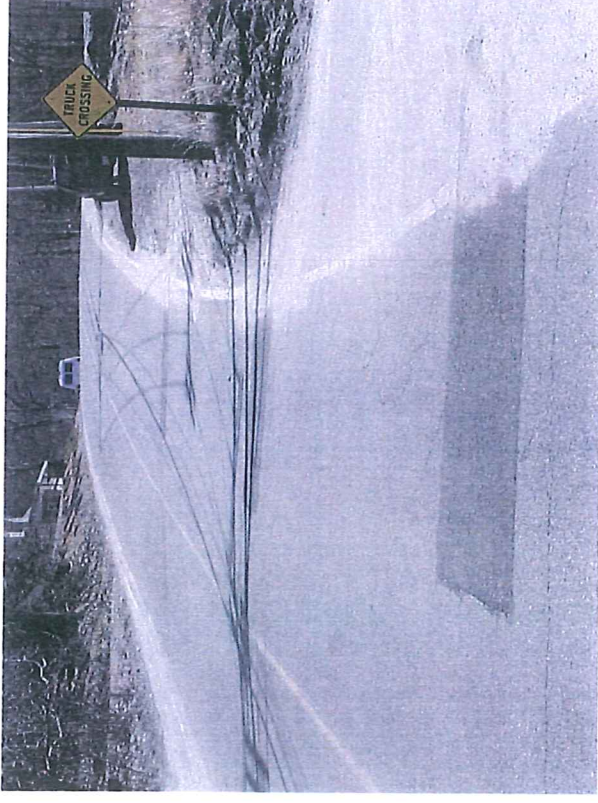
Random/Longitudinal Cracks

Random cracks are transverse cracks caused by seasonal elongation and contraction of the pavement. These will occur within a few years of any new bituminous pavement.



Utility Cuts/Other Patches

These are sections of pavement that have been repaired with new bituminous concrete pavement. (i.e., small gas/water leaks)



Raveling, Shoving, Pushing

Raveling occurs when sections of the top pavement layer become loose and separate from the binder. It indicates deteriorating bond between pavement layers.



Main Line Trench

Main line trenches are longitudinal sections of pavement that have been repaired with new bituminous concrete pavement usually after a main line utility repair.



Potholes

A crater like hole of varying size caused by freezing and thawing of the road surface particularly in areas of other defects, cracks, etc.



Rutting

Longitudinal depressions, “wheel ruts”, exist in travel lanes or edge of roadway. Due primarily to repetitive pressure along one line of travel causing improper sub base to compact and/or move laterally.



Alligator Cracks

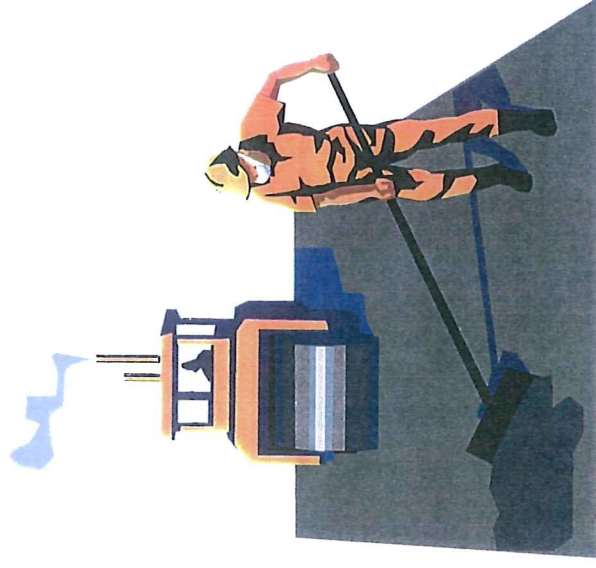
Also called
“spider webbing”
& often found at
or around
manholes, catch
basins, and/or
driveways where
they meet roads.



FUNDING

The DPW resurfacing program is predicated on the State maintaining current levels of Chapter 90 funding.

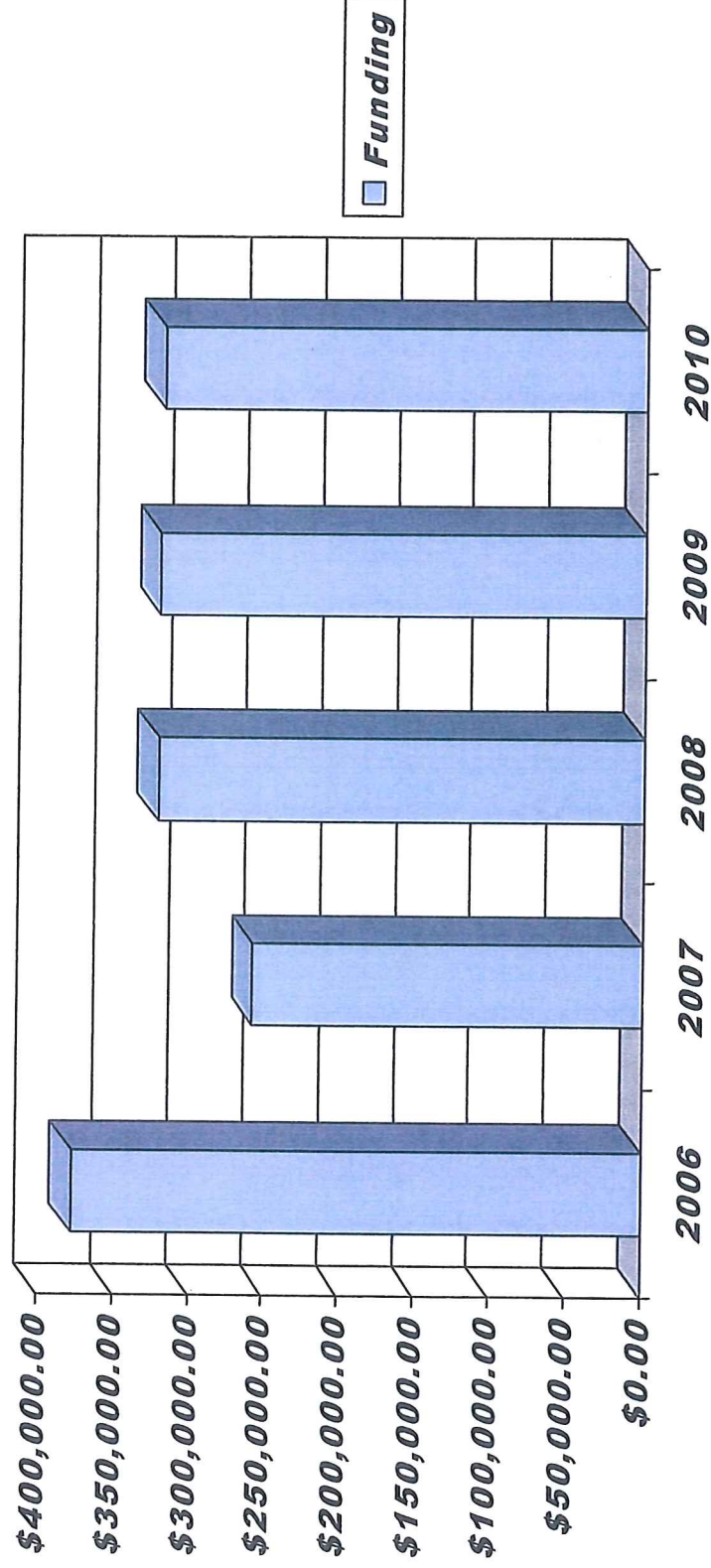
\$319,828.00 for FY10



Chapter 90 State Aid Funding History

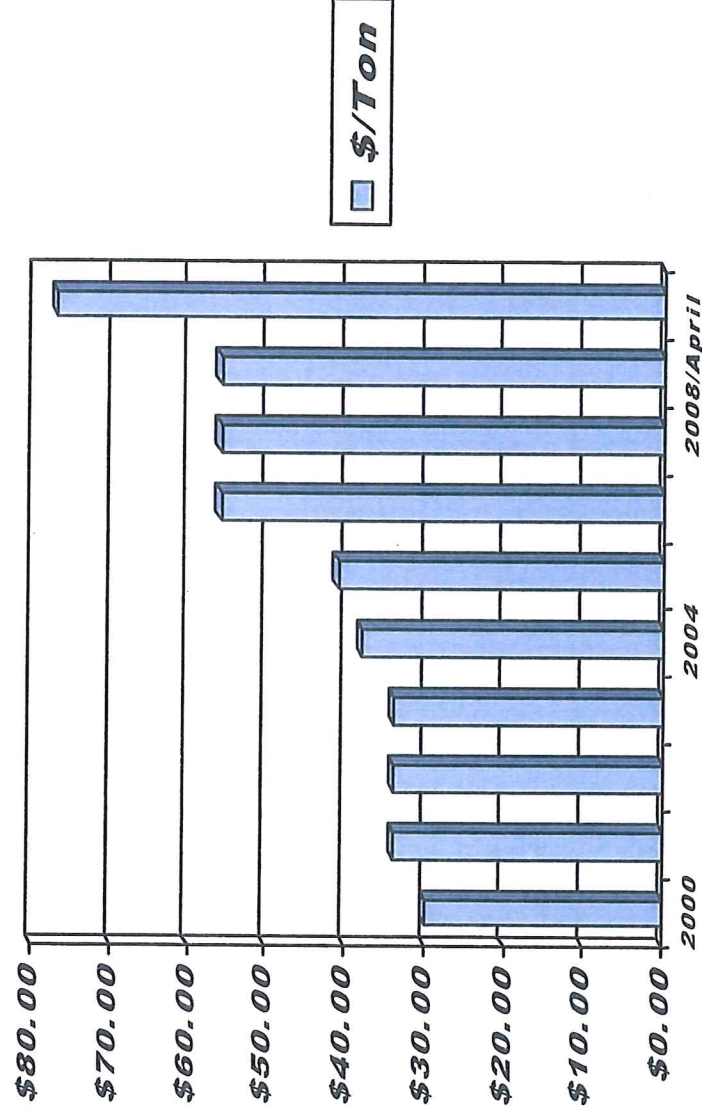
Chapt. – 291B & Sup.	\$377,704.00
Chapt. – 291C	\$258,031.00
Chapt. – 291D	\$322,432.00
Chapt. – 291D 50865	\$321,809.00
FY10 – 50865 (No longer individual chapters)	\$319,828.00

Graph of Lunenburg's Chapter 90 Funding

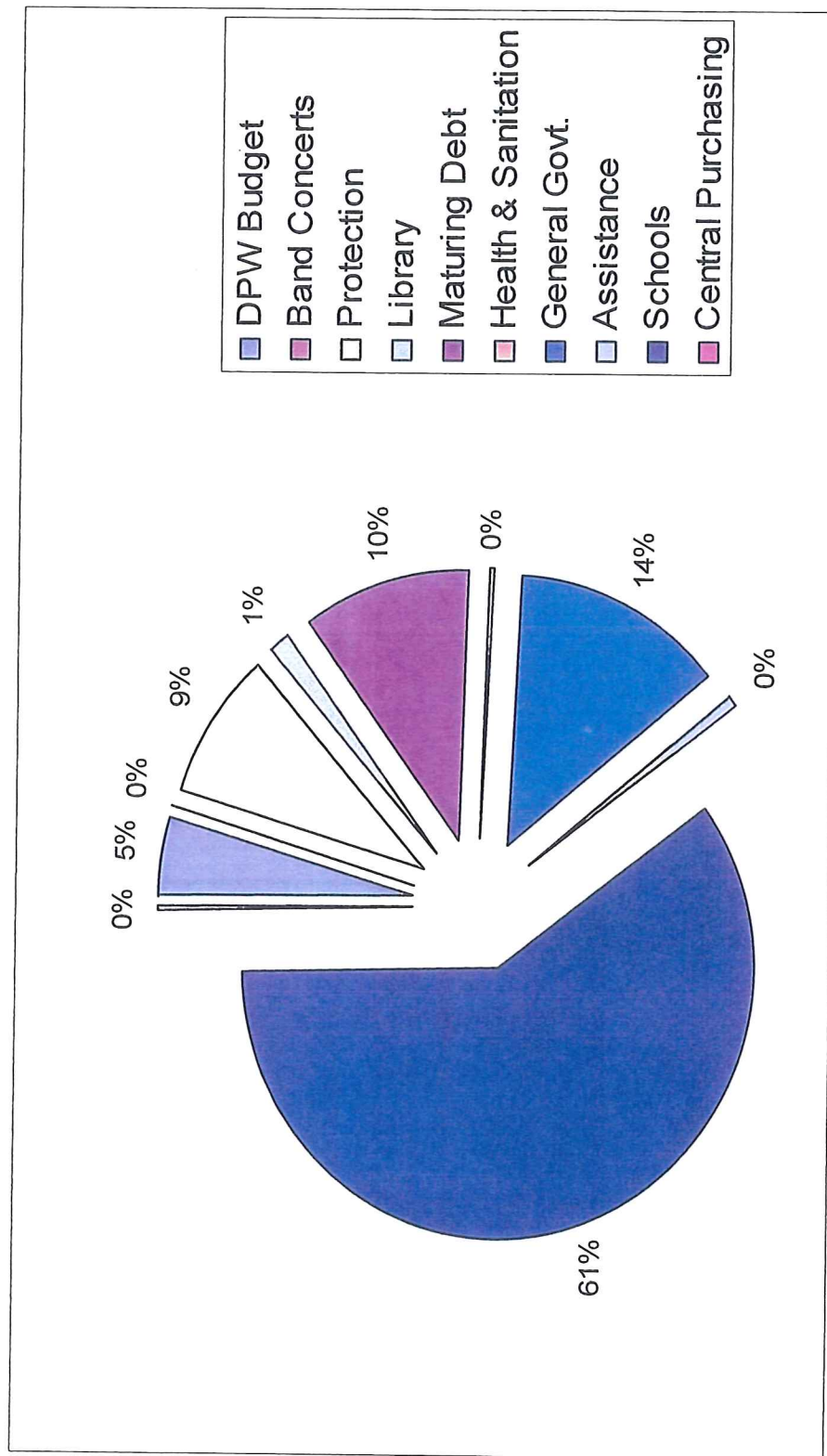


Asphalt Prices per Ton – 10 Year History

2000	\$30.00
2001	\$34.00
2002	\$34.00
2003	\$34.00
2004	\$38.00
2005	\$41.00
2006	\$56.00
2007	\$56.00
2008 (April)	\$56.00
2008 (August)	\$77.00



Percentage of Omnibus Budget Allocated for DPW FY09



Percentage of DPW Budget allocated for paving

